

Questions

MathonGo

Q1 - 2024 (05 Apr Shift 1)

Let the line $2x + 3y - k = 0$, $k > 0$, intersect the x -axis and y -axis at the points A and B, respectively. If the equation of the circle having the line segment AB as a diameter is $x^2 + y^2 - 3x - 2y = 0$ and the length of the latus rectum of the ellipse $x^2 + 9y^2 = k^2$ is $\frac{m}{n}$, where m and n are coprime, then $2m + n$ is equal to

(1) 11

(2) 10

(3) 12

(4) 13

Q2 - 2024 (09 Apr Shift 1)

Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = f \circ g(10)$, $b = g \circ f(3)$. If e and l denote the eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then $8e^2 + l^2$ is equal to.

(1) 8

(2) 16

(3) 6

(4) 12

Questions

MathonGo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

Answer Key

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

Q1 (1) mathongo

// mathongo

Q2 (1) mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

// mathongo

Solutions

MathonGo

Q1

Centre of the circle = $\left(\frac{3}{2}, 1\right)$

Equation of diameter = $2x + 3y - k = 0$

$$2\left(\frac{3}{2}\right) + 3(1) - k = 0$$

$$\Rightarrow k = 6$$

Now, Equation of ellipse becomes

$$x^2 + 9y^2 = 36$$

$$\frac{x^2}{6^2} + \frac{y^2}{2^2} = 1$$

$$\text{length of } LR = \frac{2b^2}{a} = \frac{2 \cdot 2^2}{6} = \frac{8}{6} = \frac{4}{3} = \frac{m}{n}$$

$$\therefore 2m + n = 2(4) + 3 = 11$$

Q2

$$f(x) = x^2 + 9 \quad g(x) = \frac{x}{x-9}$$

$$a = f(g(10)) = f\left(\frac{10}{10-9}\right)$$

$$= f(10) = 109$$

$$b = g(f(3)) = g(9+9)$$

$$= g(18) = \frac{18}{9} = 2$$

$$E: \frac{x^2}{109} + \frac{y^2}{2} = 1$$

$$e^2 = 1 - \frac{2}{109} = \frac{107}{109}$$

$$\ell = \frac{2(2)}{\sqrt{109}} = \frac{4}{\sqrt{109}}$$

$$8e^2 + \ell^2 = \frac{8(107)}{109} + \frac{16}{109}$$

$$= 8$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)